Technical notes:

- These bullet points vary between direct citations from the papers and paraphrases
- The summaries are extensive (maybe verbose), so I have highlighted what I believe to be the most important points or elements

General content/conceptual notes:

- GPT2 seems to be one of the most popular LMs for generating text
- We need to consider which of these papers (and others) would be most pertinent to our project

- **"Through the Looking Glass: Learning to Attribute Synthetic Text Generated by Language Models," Munir et al (2021)**
- https://aclanthology.org/2021.eacl-main.155/
  - **Attribution**
  - **Primary research question:** how do different attributors perform when tracing synthetic text back to the parent LM?
  - Uses "adversary" as a term to describe the agent who actively generates synthetic text; LMs are "threat models"
  - ML approach that aims to attribute bodies of synthetically-generated text to their source LMs
  - Key insight that texts produced by different LMs have subtle differences
  - Evaluates a series attributors on a corpus of synthetic texts
  - Texts are generated in a controlled manner using a combination of LMs, sampling parameters, and fine-tuning
  - **Pre-trained and fine-tuned LMs**
    - **Four pre-trained LMs:** Open AI GPT, Open AI GPT2, XLNet, and BART
      - XLNet and BART are both based on the BERT architecture, but outperform BERT due to their respective differences
        - XLNet uses generalized autoregressive pretraining; BART combines the bidirectional encoder of BERT with the autoregressive decoder used by GPT

        - GPT and GPT2 have identical architecture, and only differ in terms of the amount of data they were trained on
    - **The "adversaries/threat models" then fine-tune the pre-trained LMs**
    - **One fine-tuned LM:** Open AI GPT2
        - Only GPT2 is the only fine-tuned model used for text generation due to its success in the area (based on previous research)
        - Reddit data used for fine-tuning
            - r/relationships, r/conspiracy, r/technology, and r/changemyview
            - Chosen based on qualitative differences between the subreddits' common words
    - **Generating the text**
        - LMs generate text based on an input prompt from a randomly-selected Reddit submission from one of the four aforementioned subreddits
        - Three different lengths
            - Short (<= 40 words)
            - Medium (40-100 words)
            - Long (> 100 words)
        - 10,000 samples of text for 35 distinct domains -> 350,000 unique synthetic documents
- **Text generation parameters**
    - **Three key values** for generating text: p, k, and temperature
    - **Choose p-sampling or k-sampling**
        - "Top-p sampling" (choose from set of words with a cumulative probability that exceeds p) and "top-k sampling" (randomly chooses from top k words)
        - Study uses different k values for top-k sampling, but most successful with top-p sampling where p = 0.9
    - **Temperature control**

        - Controls likelihood of low probability words appearing in the final pool of words used for random selection
        - Previous work has shown that text generation quality suffers when temperature < 1
        - Study uses temperature = 1

#	Parameter	Values
1	Architecture	GPT, **GPT2**, XLNet, BART
2	Text Length	Short, **Medium**, Long
3	Fine-tuning Topic	**r/changemyview**, r/technology, r/relationships, r/conspiracy
4	p-value	**0.9**, 0.92, 0.94, 0.96, 0.98, 1.0
5	Temperature	0.1, 0.5, **1**, 1.5, 2
6	k-value	1, 20, 40, 80, 160

Table 1: The parameters explored (defaults in bold)

- **Attributors**
    - **Goal is to attribute a text to its pre-trained parent LM**
    - **Six attributors**
        - Decision tree classifier with Writeprints feature set (DT)
        - CNN with GloVe embeddings as its feature set (CNN)
        - Softmax classifier on pre-trained XLNet (XLNet-PT)
        - Softmax classifier on pre-trained GPT2 (GPT2-PT)
        - Softmax classifier on fine-tuned XLNet (XLNet-FT)
        - Softmax classifier on fine-tuned GPT2 (GPT2-FT)
- **Findings**
    - **Best performance**
        - XLNet-FT
            - Highest accuracy for all generated text lengths
            - Highest accuracy for attributing text for both pre-trained and fine-tuned models
            - Outperformed other approaches, which used stylometric differences and static and dynamic embeddings

Synthetic Text Length	DT (Writeprints)	CNN (GloVe)	XLNet-PT	XLNet-FT	GPT2-PT	GPT2-FT
Short (Upto 40 Words)	82	68	85	**91**	72	74
Medium (40 to 100 words)	86	73	90	**96**	71	72
Long (Above 100 words)	93	83	95	**98**	72	72

Table 3: Accuracy percentages for attributing source pre-trained LMs. Datasets contain synthetic texts of different lengths generated using pre-trained BERT, GPT, GPT2 and XLNet.

Training data includes	Test data includes	DT (Writeprints)	CNN (GloVe)	XLNet-PT	XLNet-FT	GPT2-PT	GPT2-FT
GPT2 (PT)	GPT2 (FT-r/changemyview)	78	64	86	**93**	70	71
GPT2 (FT-r/relationships)	GPT2 (FT-r/changemyview)	77	67	86	**91**	70	70

Table 4: Accuracy percentages in attributing source pre-trained LM when adversary generates synthetic text using a fine-tuned LM. In addition to GPT2 variants mentioned in columns 1 and 2, training and testing data also includes classes representing XLNet, BART, and GPT.

- **"Cross-Domain Detection of GPT-2-Generated Technical Text," Rodriguez et al (2022)**
- https://aclanthology.org/2022.naacl-main.88/
  - **Attribution and detection**
  - **Aims to address issues with synthetic text generation in contemporary research**
    - Most work has focused on web text or news, rather than domain-specific technical text
      - Tackling domain-specific technical text can be done by subject matter experts (SMEs)
      - However, SMEs are a limited resource: it is difficult for non-domain evaluators to accomplish the same task
      - Previous work has also focused on relatively short text samples, rather than full-length technical papers
  - **Primary research question:** can a synthetic text detector work for text in different domains?
  - **Discussion of previous research**
    - Automatic text detection usually uses GPT2

- Similar studies (to this one) that look at cross-domain settings where the distribution of synthetic text used to train a detector differs from the target distribution
    - However, the difference in domain shift may be a product of several different factors (model architecture, decoding strategies, model size, etc.)
- References Munir et al:
    - "Munir et al. (2021) show that XLNet can accurately attribute synthetic text even when the GPT-2-generated portions of the training and test sets come from GPT-2 fine-tuned on different subreddits."
- General finding that attribution is more difficult than detection

- **Experimental paradigm**
    - Tests synthetic text from GPT2 (text is generated by the "adversary") and a detector (the human creator of which is the "defender")
    - Specifically, GPT2 medium (355 million parameters)
    - GPT2 is "domain-tuned" using real text, and generates a synthetic output
        - Technical text is used for this (primarily biomedical abstract data)
    - The output and a sample of real text are labeled by SMEs before they are fed into the detector
    - A "proxy generator" is used to mimic the (theoretically unknown) adversary's generator
        - This is under the assumption that the defender has access to few (if any) examples of the adversary's text, and no access to the adversary's generator
        - The defender then uses the out-of-domain proxy synthetic text and in-domain SME-labeled real text to train a detector model
            - Pipelined approach by first task-tuning on the proxy real and synthetic text to produce a proxy detector, before task-tuning a second time on the SME-labeled text
            - Task-tuning twice may mitigate the effects of domain shift

- Assuming that the adversary and the defender have the models of the same size (but domain-tuned on different datasets)
- Assuming a temperature value of 1.0 for the target generator and 0.8 for the proxy generator
- Assuming nucleus sampling for both the adversary and the defender (similar in success to top-k sampling for related tasks)

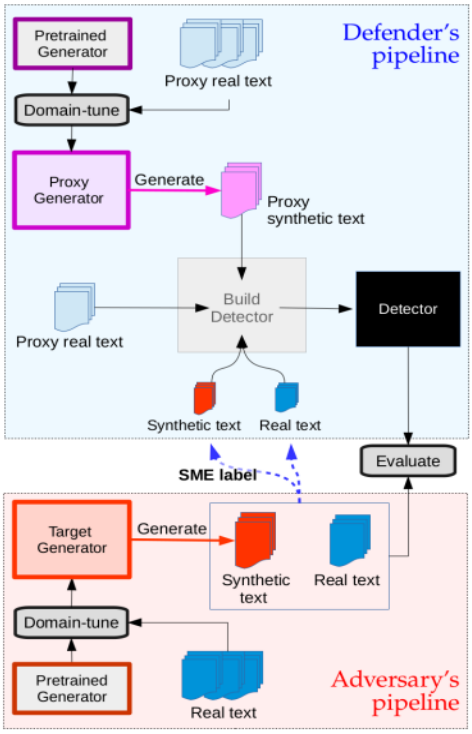

Figure 2: Experimental setup for detecting synthetic text used in this paper. Since the defender does not have access to the adversary's text generator, they develop a *proxy* generator, in order to mimic the adversary's generator as closely as possible. Proxy and target corpora will generally come from different domains.

- Detector based on a domain-tuned version of RoBERTa (mainly "RoBERTa-large-STEM") in different sizes is used to detect GPT2's synthetically-generated text
  - Proxy domain data from biomedical abstracts, physics abstracts, and biomedical-physics abstracts
- In addition to detecting synthetic documents, the experiment also examines "modified" or "tampered" documents, in which part of a body of real text is replaced with synthetic text

- This is done through conditioning the paragraphs, so they stay thematically and stylistically consistent
- Different metrics to evaluate the degree of modification in a document (S1-T exists to combat high levels of false-positives for short paragraphs in S1 and S2)
- Also, ranking of modified documents based on how much of their content is synthetic

- **(S1)** Is at least one paragraph in the document generated? The probability that a document with paragraph scores $\{s_i\}$ has at least one synthetic paragraph is then given by:

$$P = 1 - \prod (1 - s_i)$$

- **(S2)** What fraction of a document is generated? For a document with N paragraphs, this is:

$$F = \frac{1}{N} \sum \mathbb{1}_{s_i > 0.5}$$

- **(S1-T)** Is at least one paragraph p_i longer than a given threshold T synthetic? The probability that this is the case is:

$$P = 1 - \prod (1 - s_i \cdot \mathbb{1}_{\text{len}(p_i) > T})$$

- Findings
  - RoBERTa-based detectors can successfully be adapted from one scientific discipline to another with relatively small amounts of in-domain labeled data
  - RoBERTa-large generally has a higher accuracy than RoBERTa-base in most cases and always a higher precision than RoBERTa-base
    - Implying that model size plays a role in how well an LM can detect synthetic text
  - Pre-training RoBERTa on a diverse corpus of STEM technical text improved performance in most cases

- RoBERTa-large-STEM outperformed RoBERTa-large even when the target and proxy domains were close (ex. using biomedical proxy data)
- Accuracy increases as the number of SME-labeled samples and proxy samples increases when task-tuning RoBERTa-large

SME samples	Proxy samples: 0	100	500	1k	10k
0		.60	.78	.84	.93
100	.77	.82	.87	.89	.94
500	.85	.89	.90	.90	.95
1k	.91	.91	.90	.92	.94
10k	.96	.95	.96	.96	.96

Table 5: Accuracy when task-tuning RoBERTa-large on conditioned proxy data, evaluated on a balanced subset of 1000 real and 1000 synthetic paragraphs.

- **Check out these papers as well:**
  - "Prevalence of nonsensical algorithmically generated papers in the scientific literature," Cabanac and Labbé (2021)
    - https://asistdl.onlinelibrary.wiley.com/doi/10.1002/asi.24495
  - "A Benchmark Corpus for the Detection of Automatically Generated Text in Academic Publications," Liyanage, Buscaldi, and Nazarenko (2022)
    - https://aclanthology.org/2022.lrec-1.501.pdf
  - "Synthetic Text Detection: Systemic Literature Review," Guerrero and Alsmadi (2022)
    - https://arxiv.org/abs/2210.06336